

W812 Docking Document (MQTT Method)

1.Equipment Us

Users can use our **multicast scanning tool** to configure IP information (including: IP, port, subnet mask, gateway, preferred DNS, and alternative).

DNS) and MQTT information (including: server address, port, account number, password, and guaranteed time) can be used.

2. Modify attributes (type: 2)

1) Modify the IP information in the following format.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/attrs
```

Message body (list only the key fields).

```
1      {
2      "type" : 2,
3      "from" : "server serial number",
4      "to" : "Gateway Device Serial Number",
5      "timestamp" : 1596070698597, // UTC timestamp
6      "payload" : {
7      "network" : {
8          "eth0" : {
9              "ip" : "192.168.123.123", // IP address
10             "subnetmask" : "255.255.250.0", // subnet mask
11             "gateway" : "192.168.123.1", # gateway address
12             "dns_prefer" : "223.5.5.5", // Preferred DNS
13             "dns_alter" : "223.6.6.6" // alternate DNS
14         }
15     }
16 }
17 }
```

2) Modify the MQTT message in the following format.

Topic.

```
1 /voerka/0/devices/< gateway device serial number>/attrs
```

Message body (list only the key fields).

```
1 {
2     "type" : 2,
3     "from" : "Server Serial Number",
4     "to" : "Gateway Device Serial Number",
5     "timestamp" : 1596070698597, // UTC timestamp
6     "payload" : {
7         "mqtt" : {
8             "broker" : "tcp://192.168.116.123:188", // MQTT Server
                                                    Address
9             "username" : "admin", // username
10            "password" : "123456", // password
11            "keepalive" : 30, // heartbeat interval
12            "reconnectPeriod" : 4, // reconnect
interval
13            "connectTimeout" : 15 // Connection
timeout
14        }
    }
```

3. Action (type: 4)

1) Wireless watch: text message delivery

Send down text messages to the wireless watch, up to 84 bytes of content, **GB2312** characters, supporting simplified **Chinese characters (Chinese GB2312)**, English, Japanese, Arabic numerals, punctuation marks. As the transmitting module needs some time to process the message, the sending interval needs more than **3 seconds**, the gateway will do the queuing process and store up to 50 pieces of data. The format is as follows.

Topic.

```
1 /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1 {
2     "type" : 4,
3     "from" : "Server Serial Number",
4     "to" : "Gateway Device Serial Number",
5     "timestamp" : 1596070698597, #UTC timestamp
6     "payload" : {
7         "action" : "wireless_watch_text",
8         "address" : "192.168.1.100", # watch address destination address can be
watch address, group address or broadcast
9         "content" : "Welcome to the w812 Smart Gateway" # Text content, up to 84
ASCII,
Chinese GB2312
11    }
```


2) Wireless watch: time synchronization

The watch time is synchronized according to the **timestamp** field and supports point-to-point and broadcast delivery formats as follows.

Topic.

```
1 /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1 {
2     "type" : 4,
3     "from" : " Server Serial Number",
4     "to" : "Gateway Device Serial Number",
5     "timestamp" : 1596070698597, #UTC timestamp
6     "payload" : {
7         "action" : "wireless_watch_time",
8 "address" : "192.168.1.100", # watch address The destination address can be a watch address or
    a broadcast address (25)
9 "year": "2022",#Year four digits
10     "month": "1",#month
11     "day": "1",#date
12     "hour": "13",#hour24hour system
13     "minute": "0",#minutes
14     "second": "0",#seconds
15     "week": "1" #Value 1 means Monday Range(1~7)
16     }
17 }
```

3) Wireless Watch: Clear Message History

The format is as follows.

Topic.

```
1 /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1      {
2          "type" : 4,
3          "from" : "server serial number",
4          "to" : "Gateway Device Serial Number",
5          "timestamp" : 1596070698597, #UTC timestamp
6          "payload" : {
7              "action" : "wireless_watch_reset_history",
8              "address" : "192.168.1.100" # watch address destination address can be a watch
address, group address or broadcast
9          }
10     }
```

4) Wireless Watch: Clear Message History

The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1      {
2          "type" : 4,
3          "from" : "Server Serial Number",
4          "to" : "Gateway Device Serial Number",
5          "timestamp" : 1596070698597, #UTC timestamp
6          "payload" : {
7              "action" : "wireless_watch_reset_history",
8              "address" : "192.168.1.100" # watch address destination address can be a
watch address, group address or broadcast
9          }
10     }
```

5) Wireless watches: Configuring watch parameters

The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1      {
2          "type" : 4,
3          "from" : " server serial number",
4          "to" : " Gateway Device Serial Number",
5          "timestamp" : 1596070698597, #UTC timestamp
6          "payload" : {
7              "action" : "wireless_watch_config",
8              "address" : "192.168.1.100", # watch address
9              "frequency": "434.00", # transmitter frequency, take the value range
10             420.00~450.00MHz
11             "network_id": "0",#network_id,range 0~255
12         "group" : {"0": "1.1.1.1", "1":"2.2.2.2"...}          # groups of addresses, up to10 groups (corresponding to
13                                     0~9)
14     }
```

Wireless watch check whether the watch is online

The format is as follows.

Topic :

/voerka/0/devices/<Gateway Device Serial Number
>/actions

Message body (list only the key fields) :

```
{
    "type" : 4,
    "from" : "server serial number",
    "to" : "Gateway Device Serial Number",
    "timestamp" : 1596070698597, #UTC timestamp
    "payload" : {
        "action" : "wireless_watch_online",
        "address" : "192.168.1.100" # Watch mac address
    }
}
```

```
}
```

```
...
```

Wireless watch reporting device is online

The format is as follows:

Topic : /voerka/0/devices/<Gateway Device Serial Number
>/status/online

```
...
```

Message body (list only the key fields) :

```
{
```

"from" : "faff3abb8634/4219664151",#Here is the gateway address/watch address (watch address
to Long integer)

```
"payload" : {
```

```
  "code" : 1001,
```

```
  "level" : 1,
```

```
  "status" : {
```

```
    "online" : true      #Indicates the device online
```

```
  }
```

```
},
```

```
"timestamp" : 3933773,
```

```
"to" : "",
```

```
"type" : 6
```

```
}
```

```
...
```

Wireless watch response is received online

If no reply message is received, the message reported by the device is sent repeatedly (five times per
1s)

The format is as follows:

Topic:

/voerka/0/devices/<Gateway Device Serial Number
>/status/online

...

Message body (list only the key fields):

```
{
  "from" : "faff3abb8631", #server serial number

  "payload" : {
    "code" : 200
  },
  "timestamp" : 4033615,
  "to" : "faff3abb8634/4219664151",#Here is the gateway address/watch address (watch address to
Long integer)
  "type" : 6
}
```

6) W812 Gateway: Frequency value and network ID number modification

The format is as follows.

Topic.

1 /voerka/0/devices/< gateway device serial number>/actions

Message body (list only the key fields).

```
1      {
2          "type" : 4,
3          "from" : "server serial number",
4          "to" : "Gateway Device Serial Number",
5          "timestamp" : 1596070698597, #UTC timestamp
6          "payload" : {
7              "action" : "wireless_watch_config",
8              "frequency": "434.00", # transmitter frequency, take the value range
420.00~450.00MHz
```

```
9      "network_id": "0",#network_id,range 0~255
10      }
11  }
```

6) W812 Gateway: Upgrade

When the equipment needs to be upgraded, our company will provide the remote upgrade package (zip format), the customer needs to unzip it and extract the upgrade file when sending the upgrade instruction.

xxx.system (or possibly xxxx.full) and the MD5 checksum. Currently, only HTTP protocol is supported to download the upgrade package. The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/actions
```

Message body (list only the key fields).

```
1      {
2      "type" : 4,
3      "from" : " Server Serial Number",
4      "to" : "Gateway Device Serial Number",
5      "timestamp" : 1596070698597, // UTC timestamp
6      "payload" : {
7      "action" : "upgrade",
8      "url" : "http://xxxxx.system", // URL of the upgrade package
9      "md5" : "a1f984586aa73d891b7927fdaa555f3f" // MD5 checksum of xxxx.system
10     }
11     }
```

4. Event (type: 6)

1) Pager event triggering

An event is triggered each time the pager button is clicked. The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/events
```

Message body (list only the key fields).

```
1      {
2      "type": 6,
3      "timestamp": 1596010041840, // UTC timestamp
4      "from": " Gateway Device Serial Number",
5      "payload": {
6      "id": "123456", // Pager ID
7      "key": "1" // pager key value
8      }
9     }
```

2) Online status reporting

The device will set up a legacy message (sent by MQTT on behalf of the device when it is offline) at the same time when registering for MQTT, and will send an online message when the registration for MQTT is successful. The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/status/online
```

Message body (list only the key fields).

```
1      {
2      "type": 6,
3      "timestamp": 1596010041840, // UTC timestamp
4      "from": "Gateway Device Serial Number",
5      "payload": {
6      "status": {
7      "online": true // online true, offline false
8      }
9      }
10     }
```

5. Device time synchronization (type: 11)

Since the device does not have a built-in RTC module, the time needs to be synchronized from the "timestamp" field of the MQTT message. The device will synchronize the time from any MQTT message it receives. The server can synchronize the device time at regular intervals after the device is online to prevent inaccurate timestamps reported by the device. If only the device's time is synchronized, send the following data. The format is as follows.

Topic.

```
1      /voerka/0/devices/< gateway device serial number>/answer
```

Message body (list only the key fields).

```
1      {
2      "type": 11,
3      "timestamp": 1596010041840, // UTC timestamp
4      "from": "server-serial-number",
5      "to": "Gateway Device Serial Number",
6      "payload": {
7      }
8      }
```

